Tiered Memory Management Beyond Hotness

CS-471: Project Status Report

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1 Research Problem

[TODO: Remind the problem in a few lines: placement based on hotness ignores memory-level parallelism, and the goal of reproducing AOL and SOAR.]

[TODO: State whether the main focus has changed since the proposal.]

2 Questions, Hypotheses, and Tests

2.1 Methodology and Testing Environment Updates

[TODO: Machine actually used and how it differs from the assumptions of the proposal and from the paper.]

[TODO: Changes to the methodology: counters used, measurement procedure, how the slow tier is emulated.]

2.2 Current Status of Research Questions

2.2.1 Q1: Can hotness be a faithful metric to measure performance?

Insight: [TODO: I1 from the proposal.]
Experiment: [TODO: E1, and the test of H1′.]
Status: [TODO: Done, partial or not started.]

2.2.2 Q2: If not, how can a better metric be defined?

Insight: [TODO: I2 from the proposal.]
Experiment: [TODO: E2, and the evaluation of H2′.]
Status: [TODO: Done, partial or not started.]

2.2.3 Q3: With this metric, can we improve first data placement (SOAR)?

Insight: [TODO: I3 from the proposal.]
Experiment: [TODO: E3.]
Status: [TODO: Done, partial or not started.]

2.2.4 Q4: Can these elements improve PolyRack?

Insight: [TODO: I4 from the proposal.]
Experiment: [TODO: E4.]
Status: [TODO: Done, partial, not started, or dropped.]

2.3 Anomalies and Updates to the Hypothesis Tree

[TODO: Any anomaly observed, and whether the hypothesis tree changes. Include the updated figure if it does.]

3 Progress, Results, and Analysis

3.1 Experimental Setup

[TODO: What has been brought up and verified on the machine.]

3.2 E1: Hotness versus Criticality

[TODO: Configurations, number of runs, results and conclusion on H1.]



Figure 1: [TODO: Caption.]

3.3 E2: Slowdown Predictors (Preliminary)

[TODO: Calibration of a, b and comparison of the predictors.]

Predictor	Pearson	MAE
P	[TODO: –]	[TODO: –]
$P \times K$	[TODO: –]	[TODO: –]
Stalls per miss (H2′)	[TODO: –]	[TODO: –]
Hotness	[TODO: –]	[TODO: –]

Table 1: [TODO: Caption.]

3.4 SOAR: Feasibility

[TODO: What the authors' artifact requires, what is needed to run it on our machine, and any first result.]

4 Challenges

4.1 Hardware Differences

[TODO: Challenges coming from the CPU and its counters.]

4.2 Tiered Memory Emulation and Baselines

[TODO: Challenges with the slow tier, memory tiers and custom kernels.]

4.3 Authors' Artifact

[TODO: Challenges in porting the code.]

4.4 Help Needed from the Teaching Team

[TODO: Concrete requests.]

5 Plan

[TODO: Say whether the project is on schedule with respect to the proposal.]
[TODO: Division of the work between the team members.]

Date	Milestone	Status
[TODO: date]	[TODO: E2 completed]	[TODO: –]
[TODO: date]	[TODO: SOAR results]	[TODO: –]
Nov. 27	[TODO: E3 on selected workloads and ratios]	[TODO: –]
Dec. 4	[TODO: (if time permits) E4 on PolyRack]	[TODO: –]
Dec. 10	Poster deadline	
Dec. 17	Poster session	
Dec. 20	Final report deadline	

Table 2: Updated milestones.